

Vamshi Krishna Nayini

Senior Data Engineer

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PROFESSIONAL SUMMARY:

- ✓ Data Engineer with more than **10+ years** of experience in multiple technology methodologies like Azure Services, and Big Data, including Analysis, Design, and Development of Big Data using Hadoop, Python, Data Lake, Scala, and PySpark, and database and data warehousing development using My SQL, Oracle, and Data warehouse.
- ✓ 5 years of experience as Azure Cloud Data Engineer in Microsoft Azure Cloud technologies including Azure Data Factory (ADF), Azure Data Lake Storage (ADLS), Azure Synapse Analytics (SQL Data warehouse), Azure SQL Database, Azure Analytical services, Azure Cosmos NoSQL DB, Azure Key vaults, Azure DevOps, Azure HDInsight Big Data Technologies like Hadoop, Apache Spark, and Azure Data bricks.
- ✓ 4 years of experience as a Data warehouse developer handling Microsoft Business Intelligence Tools.
- ✓ Hands-on experience in Azure Cloud Services (PaaS & IaaS), Azure Synapse Analytics, SQL Azure, Data Factory, Azure Analysis Services, Application Insights, Azure Monitoring, Key Vault, and Azure Data Lake.
- ✓ Experience in developing pipelines in Spark using Scala and PySpark.
- ✓ Managed and administered Hadoop clusters, ensuring high availability, scalability, and optimal performance.
- ✓ Experience building ETL (Azure Data Bricks) data pipelines leveraging PySpark, and Spark SQL and extensively worked on Azure Databricks.
- ✓ Proficient in Azure Data Factory to perform Incremental Loads from Azure SQL DB to Azure Synapse.
- ✓ Experience in building the Orchestration on Azure Data Factory for scheduling purposes.
- ✓ Designed and optimized CDC pipelines to efficiently capture and replicate data changes from various source systems such as relational databases (e.g., MySQL, PostgreSQL, Oracle) and NoSQL databases (e.g., MongoDB, Cassandra).
- ✓ Hands-on experience with cloud data storage and analytics platforms, including GCP Cloud Storage, Azure Data Lake Storage (ADLS Gen2), and BigQuery.
- ✓ Integrated Hadoop with various components of the big data ecosystem, including HBase, Hive, Spark, and Pig, to support diverse data processing and analytics requirements.
- ✓ Implemented indexing and partitioning strategies for large datasets to enhance query performance.
- ✓ Orchestrated data integration pipelines in ADF using various Activities like Get Metadata, Lookup, For Each, Wait, Execute Pipeline, Set Variable, Filter, until, etc.
- ✓ Strong experience in migrating other databases to Snowflake.
- ✓ Experience in using Snowflake Clone and Time Travel.
- ✓ Experience with Snowflake Multi-Cluster Warehouses.
- ✓ Experience in using Star Schema and Snowflake schema for Modeling and using FACT & Dimensions tables, Physical & Logical Data Modeling.
- ✓ Proficient in integrating third-party platforms like Google Marketing Platform and Salesforce MCI with Azure architectures to provide holistic business insights and support cross-functional, data-driven decision-making.
- ✓ Experienced in developing custom connectors for smooth ingestion of marketing metrics and campaign data from diverse sources, enabling detailed reporting and performance optimization.

- ✓ Skilled at designing impactful Power BI dashboards that consolidate marketing and operational data, empowering stakeholders with actionable insights to assess campaign effectiveness and ROI.
- ✓ Collaborated with data scientists to identify opportunities for automation and process improvement.
- ✓ Hands-on working experience and developing large-scale data pipelines using spark and hive.
- ✓ Experience working with ARM templates to deploy in production using Azure DevOps.
- ✓ Experience in developing very complex mappings, reusable transformations, sessions, and workflows using the Informatica ETL tool to extract data from various sources and load it into targets.
- ✓ Used T-SQL stored procedures to transfer data from OLTP databases to the staging area and finally transfer it into data marts and acted XML.
- ✓ Used various file formats like Avro, Parquet, Sequence, JSON, ORC, and text for loading data, parsing, gathering, and performing transformations.
- ✓ Extensive experience with T-SQL in constructing Triggers and tables, implementing stored Procedures, Functions, Views, User Profiles, Data Dictionaries, and Data Integrity.
- ✓ Hands-on experience with Confluent Kafka to load data from StreamSets directly into ADLS.
- ✓ In-depth knowledge in working with Distributed Computing Systems and parallel processing techniques to efficiently deal with Big Data.
- ✓ Designed and Implemented Hive external tables using a shared meta-store with Static & Dynamic partitioning, bucketing, and indexing.
- ✓ Experience in handling, configuration, and administration of databases like MySQL and NoSQL databases like MongoDB and Cassandra.
- ✓ Experienced in working with structured data using HiveQL, and optimizing Hive queries. Extensive hands-on experience tuning spark Jobs and spark performance tuning.
- ✓ Experience with solid capabilities in exploratory data analysis, statistical analysis, and visualization using R, Python, SQL, and Tableau.
- ✓ Running and scheduling workflows using Oozie and Zookeeper, identifying failures, and integrating, coordinating, and scheduling jobs.
- ✓ Experience in Data Modeling & Analysis using Dimensional and Relational Data Modeling.
- ✓ Maintained and administered GIT source code repository and GitHub Enterprise.
- ✓ Drove engagement with ITS Security and Infrastructure teams to ensure secure development and deployment of Azure-based data solutions.
- ✓ Extensive experience in designing and tuning OLAP and OLTP systems, ensuring efficient processing of transactional and analytical workloads for high-performance database environments.
- ✓ Created interactive dashboards and reports in Power BI, leveraging data from various sources for insightful business intelligence and reporting.
- ✓ Scheduled and monitored data workflows with Control-M and Apache Airflow for coordinated execution of complex tasks.
- ✓ Defining user stories and driving the agile board in JIRA during project execution, participating in sprint demos and retrospectives.
- ✓ Involved in all the phases of Software Development Life Cycle (Requirements Analysis, Design, Development, Testing, Deployment, and Support) and Agile methodologies.

TECHNICAL SKILLS:

Azure Services:

Azure Data Factory, Azure Data Bricks, Logic Apps, Functional App, Azure DevOps, Azure Synapse Analytics, Key Vault, Azure Event Hub, Azure

	Monitor, Azure Stream Analytics, Azure Blob Storage, Azure Functions, MS Fabric.
Databases:	Snowflake, MS SQL Server 2016/2014/2012, Azure Data Lake Storage Gen2, Azure SQL DB, MS Excel, MS Access, Oracle 11g/12c, Redshift, Azure Synapse, Cosmos DB, Mongo DB PostgreSQL, MySQL
Big Data Technologies:	MapReduce, Hive, Teg, Python, PySpark, Scala, Kafka, Spark streaming, Oozie, Sqoop, HDFS, Flume, Zookeeper
Hadoop Distribution:	Cloudera, Horton Works, Apache
Languages:	Python, SQL, PL/SQL, Java, HiveQL, Scala, Shell Scripting, Bash, PowerShell
Web Technologies:	HTML, CSS, JavaScript, XML, JSP, Restful, SOAP
Operating Systems:	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS
Build Automation tools:	Ant, Maven, Jenkins
Version Control:	GIT, GitHub, Bitbucket
Methodology:	Agile, Scrum
IDE & Build Tools, Design:	Eclipse, Visual Studio, PyCharm, IntelliJ IDEA, VS Code

EDUCATION

- ✓ Masters in Computer Science from University of the Wright State University, OH, USA
- ✓ Bachelors in Computer Science from Lingayas Vidyapeeth, Haryana, India

PROFESSIONAL EXPERIENCES

Vangurad, Malverin, PA.

June 2023-Present

Role: Senior Data Engineer

Responsibilities:

- ✓ Designed and implemented end-to-end ETL pipelines using Azure Data Factory, incorporating Copy Activities, Mapping Data Flows, and Control Flow Activities, improving data transfer efficiency.
- ✓ Configured Linked Services in Azure Data Factory for seamless connection and integration of diverse data sources, improving data flow and integration efficiency.
- ✓ Created and managed Datasets in Azure Data Factory to ensure accurate data mapping and transformation across multiple data sources and destinations.
- ✓ Configured Linked Services, created Datasets, and set up Integration Runtime in Azure Data Factory for seamless data integration across diverse environments.
- ✓ Integrated Azure Function Apps within Azure Data Factory pipelines for automated data processing, increasing system flexibility and responsiveness.
- ✓ Implemented advanced scheduling and automation using Scheduled Trigger, Tumbling Window Trigger, and Event-Based Trigger in Azure Data Factory to ensure timely ETL execution and data synchronization.
- ✓ Leveraged Tumbling Window Triggers and Linked Services to connect multiple data sources, reducing latency and processing time for large-scale integrations.
- ✓ Developed real-time data ingestion solutions by integrating Azure Data Factory with Azure Event Hubs, improving synchronization accuracy and reducing latency.
- ✓ Integrated Azure Key Vault to securely manage secrets, keys, and certificates within Azure Data Factory pipelines, significantly enhancing security and ensuring compliance with industry standards.
- ✓ Set up Azure Monitor for comprehensive monitoring of data pipelines, enabling proactive identification and resolution of performance bottlenecks.

- ✓ Hands-on experience in unified development on Fabric (SaaS) using Data Factory, Data Science, Data Warehousing, Data Engineering, Data Activator, Power BI, Real-Time Analytics, and OneLake.
- ✓ Created shared workspaces in Fabric for collaborative development environments.
- ✓ Designed and developed end-to-end solutions for data ingestion, processing, and visualization using Fabric's capabilities.
- ✓ Developed complex SQL scripts in Azure Synapse Analytics, utilizing columnstore indexes and PolyBase for external data source integration, boosting query performance.
- ✓ Leveraged the Massively Parallel Processing (MPP) architecture in Azure Synapse to handle large datasets and optimize critical reporting dashboards.
- ✓ Integrated Power BI with Azure Synapse Analytics for real-time data visualization, reducing time-to-insight by 40%.
- ✓ Implemented data integration and transformations within Azure SQL Server, ensuring high availability and performance for enterprise-level applications.
- ✓ Optimized complex T-SQL queries in Azure SQL Server for efficient data manipulation and reporting, enhancing data retrieval performance.
- ✓ Designed and developed scalable data models in Snowflake, utilizing its MPP architecture for high-performance data storage and querying.
- ✓ Employed Snowflake's Time Travel and Zero-Copy Cloning features for efficient data backup and recovery, minimizing downtime and mitigating data loss risks.
- ✓ Developed and optimized Snow SQL queries for efficient ETL operations in Snowflake.
- ✓ Designed and implemented Snowflake Schema for optimized data warehousing, enabling efficient data storage and retrieval.
- ✓ Developed Star Schema data models for OLAP systems, ensuring efficient query performance and simplified data navigation.
- ✓ Developed and maintained real-time data streaming pipelines using Apache Kafka, enabling efficient data flow for real-time analytics.
- ✓ Integrated Kafka with Azure Event Hubs for seamless data streaming between cloud and on-premises systems, enhancing data synchronization and reducing latency.
- ✓ Managed and optimized data storage and processing using file formats like PARQUET, JSON, and CSV for compatibility and efficiency across systems.
- ✓ Utilized file formats such as Parquet and ORC for efficient storage and querying in big data environments, improving data retrieval speeds.
- ✓ Designed Power BI reports and dashboards combining clinical, operational, and marketing data from Google Marketing Platform and Salesforce MCI, empowering stakeholders with actionable insights to support data-driven decision-making in patient care and business strategies.
- ✓ Created interactive Power BI dashboards to visualize key business metrics, improving decision-making and enabling real-time insights.
- ✓ Worked in an Agile environment, collaborating with cross-functional teams to deliver high-quality data solutions on time and within budget.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen2, Azure Blob, Snowflake Data Warehouse, Azure Event Hubs, Azure Functions, Azure Logic Apps, Function Apps, Azure Monitor, Azure Synapse Analytics, Azure DevOps, Azure Fabric, Azure Key Vault, MS SQL, Oracle, Python, PySpark, JIRA, Kafka, Power Bi.

Azure Data Engineer

Responsibilities:

- ✓ Designed and implemented intricate data integration workflows in Azure Data Factory, using a variety of activities such as Copy Activity, Mapping Data Flows, and Control Flow.
- ✓ Configured Linked Services to connect multiple data sources and set up Triggers for automated pipeline execution.
- ✓ Utilized Azure Data Factory's Integration Runtime to ensure secure data transfers between on-premises systems and the cloud, deploying Self-Hosted Integration Runtime for data access behind corporate firewalls.
- ✓ Streamlined data storage and management in Azure Data Lake Storage Gen2, leveraging the Hierarchical Namespace for optimized data organization.
- ✓ Utilized Azure Databricks Workspace for creating and managing clusters, developing notebooks to handle data processing and analysis tasks.
- ✓ Developed ETL pipelines using Azure Databricks, optimizing data transformations through the use of Spark DataFrames and RDDs.
- ✓ Integrated Azure Data Lake Storage Gen2 with Azure Active Directory (AAD) to ensure secure and seamless authentication for data access control.
- ✓ Built efficient data pipelines in Azure Synapse Analytics, employing Data Flows for no-code ETL and Notebooks for more complex transformations.
- ✓ Developed data processing scripts in Python, utilizing libraries such as Pandas and NumPy for effective data transformation and manipulation.
- ✓ Delivered end-to-end analytics solutions within Azure Synapse Analytics Workspace, combining SQL Pools for warehousing and Spark Pools for large-scale data processing.
- ✓ Used Azure Synapse Link to facilitate near real-time analytics by integrating with Azure Cosmos DB, ensuring synchronized operational data.
- ✓ Executed a one-time migration of multi-state data from SQL Server to Snowflake using Python and SnowSQL.
- ✓ Created data ingestion pipelines using Sqoop and Flume to import structured and unstructured data from on-prem systems into HDFS and Azure Blob Storage.
- ✓ Configured Kafka for real-time data streaming from various sources into Hadoop and Azure environments, ensuring minimal latency in data processing.
- ✓ Built complex data transformation workflows with Apache Pig, Hive, and Spark to clean and process large datasets, ensuring high data quality and consistency.
- ✓ Optimized large-scale data processing using MapReduce and YARN, significantly reducing execution times for major transformations.
- ✓ Managed large datasets within HDFS, Azure Data Lake, and Apache Cassandra, optimizing storage solutions for better performance and cost efficiency.
- ✓ Utilized Cosmos DB and HBase for NoSQL storage, enabling fast read/write operations for critical applications.
- ✓ Automated end-to-end data workflows with Oozie and Control-M, ensuring continuous data movement and processing across hybrid environments.
- ✓ Analyzed and optimized Zookeeper for distributed coordination and consistency management in Hadoop and Azure ecosystems.

- ✓ Conducted performance tuning of Spark jobs, SQL queries, and Hadoop clusters, improving data processing speed and resource utilization.
- ✓ Applied best practices for data partitioning, indexing, and caching to enhance performance in Hive queries and SQL Server databases.
- ✓ Secured data pipelines by integrating Azure Key Vault, ensuring compliant management of secrets, keys, and certificates.
- ✓ Managed and optimized storage using multiple file formats such as ORC, PARQUET, AVRO, JSON, and CSV, ensuring compatibility and efficiency across platforms.
- ✓ Collaborated with data analysts to create scalable reporting solutions using Hive and Spark, delivering actionable insights from processed data.
- ✓ Integrated Power BI with processed datasets, enabling seamless data visualization and driving data-driven decision-making across the organization.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen 2, Azure Blob, Logic Apps, Functional App, Snowflake, MS SQL, Oracle, HDFS, Hive, Sqoop, PIG, PySpark, Oozie, Zookeeper, SQL, Python, Scala, Kafka, Power Bi.

Goldman Sachs, Dallas, TX

May 2016-May 2018

Role: Big Data Developer

Responsibilities:

- ✓ Developed and fine-tuned complex MapReduce programs for handling large datasets, enhancing performance by 30%.
- ✓ Structured and maintained HDFS systems, ensuring efficient storage and retrieval of petabyte-scale datasets.
- ✓ Created and optimized Hive queries and HiveQL scripts for ETL processes, cutting query execution times by 25%.
- ✓ Managed and automated multi-step data pipelines with Apache Oozie across the Hadoop ecosystem.
- ✓ Integrated Apache Kafka with Hadoop for real-time data streaming, boosting data ingestion speeds by 40%.
- ✓ Designed and implemented Sqoop jobs for seamless data transfers between Hadoop and relational databases like MS SQL Server and MySQL.
- ✓ Developed PySpark and Scala scripts to process large-scale datasets, improving processing efficiency by 20%.
- ✓ Maximized the performance of Hadoop and Spark clusters by optimizing configurations and resource allocation, reducing job execution times by up to 40%.
- ✓ Optimized Apache Cassandra for distributed database management, ensuring high availability and scalability of data across multiple nodes.
- ✓ Deployed Apache Zookeeper for distributed coordination, enhancing reliability and fault tolerance within Hadoop clusters.
- ✓ Engineered data ingestion workflows with Apache Flume, ensuring efficient and reliable log data streaming into HDFS.
- ✓ Performed data modeling and query optimization in HBase, leading to faster data retrieval for low-latency applications.

- ✓ Utilized Control-M to schedule and monitor batch jobs, ensuring timely execution of critical data processing tasks.
- ✓ Developed Pig scripts for ETL processes, simplifying complex data transformations and strengthening data pipeline reliability.
- ✓ Designed and implemented scalable Cosmos DB solutions for globally distributed data storage, improving application performance.
- ✓ Conducted performance tuning on Hive and Spark jobs, significantly improving processing speeds and resource utilization.
- ✓ Automated cluster management tasks using YARN, reducing operational overhead and enhancing cluster efficiency.
- ✓ Utilized Python and YAML for configuration management and scripting, streamlining deployments and minimizing errors.
- ✓ Integrated Apache Spark with HDFS for distributed data processing, optimizing computational efficiency for Big Data analytics.
- ✓ Collaborated with cross-functional teams to deliver end-to-end data solutions, leveraging Hadoop and related technologies to meet business needs.
- ✓ Utilized Git and GitHub repositories to maintain the source code and enable version control

Environment: MS SQL, Oracle Sqoop, Pig, HDFS, Apache Cassandra, Oozie, Zookeeper, Control-M, Flume, Kafka, Apache Spark, Scala, Hive, Hadoop, HBase, Cosmos DB, MySQL, YAML, GIT, JIRA

Kroger, Cincinnati, OH

Feb 2014 to Apr 2016

Role: Data Warehouse Developer

Responsibilities:

- ✓ Extensive experience in Dimensional Data Modeling for Data Mart design, identifying Facts and Dimensions, and developing, fact tables, dimension tables, using Slowly Changing Dimensions (SCD)
- ✓ Expertise in developing Parameterized, Chart, Graph, Linked, Dashboard, Scorecards, Report on SSAS Cube using Drill-down, Drill-through and Cascading reports using SSRS.
- ✓ Designed and maintained SQL Server databases, ensuring optimal data storage, retrieval, and overall database performance.
- ✓ Led data modeling initiatives, handling both physical and logical database designs to meet complex business requirements.
- ✓ Integrated front-end applications with SQL Server backends, improving data flow and application functionality.
- ✓ Developed advanced T-SQL scripts, including stored procedures, triggers, indexes, and user-defined functions, to automate processes and enhance database performance.
- ✓ Managed data migration projects using DTS, ensuring smooth data transfers between servers and environments.
- ✓ Optimized T-SQL queries to improve performance, reducing execution times and enhancing system responsiveness.
- ✓ Utilized SSIS/DTS to transfer and transform data from sources like MS Excel, MS Access, and flat files into SQL Server, incorporating features like data conversion and derived columns.
- ✓ Designed and deployed complex SSIS packages for ETL processes, implementing detailed logging at both package and task levels to monitor data flow.

- ✓ Provided 24/7 production support, creating and testing SQL Server queries and Windows scripts to ensure database stability and availability.
- ✓ Designed and generated advanced reports with SQL Reporting Services, including cross-tab, drill-down, and OLAP reports, tailored to specific business needs.

Environment: SQL Server, Erwin, TOAD, Windows 2000, Oracle 9i, UNIX Shell Scripts, PL/SQL, Data Transformation Services (DTS), SQL Server Integration Services (SSIS), MS Excel, MS Access, SQL Server Reporting Services (SSRS).